

Test Plan Document

This test plan outlines the testing strategy and validation process for the backend API system. The project includes core functionalities such as authentication, user profile management, help requests, segment and review handling and admin-level operations.

It is developed using **Node.js**, **TypeScript**, **Express**, and **Prisma ORM** with **PostgreSQL** as the database.

Objectives

The goal of this testing process is to ensure:

- All API endpoints function as intended.
- Authentication and validation mechanisms work correctly.
- Database operations are reliable and consistent.
- Automated tests run seamlessly through GitHub Actions.
- Swagger documentation accurately describes all available endpoints.
- Manual testing through Postman confirms real-world API behavior.

Scope of Testing

Included in Scope

- Authentication and Authorization
- Profile management
- Segments and Reviews
- Help Requests and Responses

Excluded from Scope

- Frontend testing
- Load or performance testing (to be added in a later phase)

Test Environment

The testing was carried out in both local and continuous integration environments.

Key components include:

- Node.js and Express.js backend
- PostgreSQL database managed by Prisma ORM
- vitest for unit testing
- Swagger for documentation
- Postman for manual API verification
- GitHub Actions for automated testing and deployment checks

Test Strategy

Unit Testing

- Each controller and service was tested independently to confirm correct logic and behavior.
- Mocks were used for database interactions to isolate business logic.
- Testing covered all major services such as authentication, profiles, segments, reviews and help requests.

API Documentation

- All API routes were documented using Swagger (OpenAPI 3.0 format).
- Each endpoint includes method type, parameters, request body, and sample responses.
- The documentation serves as a complete reference for developers and testers.

Manual Testing with Postman :-

- All endpoints were tested manually using Postman collections.
- Tests included both valid and invalid scenarios to confirm accurate error handling and data validation.
- Authenticated routes were tested using valid and expired tokens to verify access control.

Automated Testing (GitHub Actions) :-

- GitHub Actions were configured to automatically run all unit tests and validation checks on each push or pull request.
- This ensures that every update to the codebase is automatically tested before merging, maintaining code quality and reliability.

Test Cases Summary

Module	Test Case	Expected Result	Status
Authentication	Register user	User created successfully	☒
Authentication	Login user	Returns valid token	☒
Authentication	Password reset	Password updated successfully	☒
Profile	Update profile	Returns updated data	☒
Segments	Fetch segments	List retrieved successfully	☒
Reviews	Create review	Review saved correctly	☒
Help Requests	Create request	Request created successfully	☒
Help Responses	Update response	Response status updated	☒